

Muhammad Afsah Mumtaz

AI Engineer | Machine Learning Engineer | LLM & Applied AI Systems

Dubai, United Arab Emirates | Open to Dubai / Abu Dhabi (on-site or hybrid) and remote worldwide

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Nationality: Pakistani | Visa: UAE Golden Visa (no sponsorship required) | Notice period: Immediate

PROFESSIONAL SUMMARY

AI / Machine Learning engineer who builds and ships **large language model (LLM) systems end to end** — fine-tuning, retrieval-augmented generation (RAG), multi-agent orchestration, evaluation, and serving (BSc Hons Information Technology student, Middlesex University Dubai). Fine-tuned a 2B LLM with QLoRA to **82% ARC-Challenge and 92% ARC-Easy — beating Llama 3.2 3B-Instruct — on a single 4 GB GPU**, and shipped a **live 11-agent LLM platform** orchestrating 11 model providers. Deep with Python, PyTorch, Hugging Face, and Ollama; active open-source AI-research contributor. Seeking an **AI Engineer or Machine Learning internship / junior role**. On-site or hybrid in Dubai / Abu Dhabi or remote worldwide; UAE Golden Visa holder — no sponsorship required, immediate start.

TECHNICAL SKILLS

AI / Machine Learning: Machine Learning, Deep Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Fine-Tuning (LoRA / QLoRA, PEFT), Retrieval-Augmented Generation (RAG), Multi-Agent Systems / Agentic AI, LLM Orchestration, Prompt Engineering, Model Evaluation & Benchmarking, Quantization (GGUF), Inference Optimization, Model Serving, Distillation

AI Frameworks & Tooling: PyTorch, Hugging Face Transformers, PEFT, Ollama, Anthropic Claude, OpenAI, Google Gemini, Groq, DeepSeek, Vector Search (HNSW), SQLite FTS5

Languages: Python, Rust, TypeScript, JavaScript, C++, SQL

Backend & APIs: FastAPI, Node.js, Express, REST API design, API integration, PostgreSQL, Redis, MongoDB, Linux, WSL2

Infra & Tooling: Docker, Git, GitHub Actions, CI/CD, PowerShell, CMake, Release Packaging

Frontend (supporting): Next.js, React, Tailwind CSS, Three.js, WebGL, GLSL

Practices: Agile, Testing & Reproducible Evaluation, Version Control, Automation, Technical Documentation

PROJECTS

AVA — Local-First LLM Fine-Tuning, Evaluation & Inference Lab

Tech: Python, PyTorch, Hugging Face Transformers, PEFT / QLoRA, RAG, Ollama, GGUF

- Engineered a local-first machine learning stack for training, evaluation, distillation, and serving of large language models on a single 4 GB VRAM laptop GPU.
- Fine-tuned Qwen3.5-2B with a 42 MB QLoRA adapter (0.58% trainable parameters) to 82.0% on ARC-Challenge, 92.0% on ARC-Easy, and 59.2% on MMLU, outperforming Llama 3.2 3B-Instruct (78.6%) on ARC-Challenge.
- Built a reproducible evaluation harness spanning 17 benchmarks (16,872 tasks); training peaked at 1.81 GB VRAM in 100 minutes, with open-sourced adapters and artifacts.

Pantheon-Trades — Multi-Agent LLM System (Live: pantheon-trades.vercel.app)

Tech: Python, FastAPI, Multi-LLM (Claude, Gemini, OpenAI, Groq, DeepSeek, +6), Next.js 14, PostgreSQL, Redis

- Designed and shipped a live system where an 11-agent large language model council deliberates over four structured rounds (opening, challenge, synthesis, blind vote) before any decision.
- Engineered a pluggable multi-LLM backend across 11 providers with self-calibrating agent weights (Platt / isotonic regression) driven by Brier scores.
- Validated with 714 Python and 65 Solidity tests and a 200-market backtest, demonstrating production-grade AI evaluation and reliability.

MALD — Local AI Knowledge System with Cited RAG

Tech: Rust, SQLite FTS5, HNSW vector indexing, RAG, Ollama, Terminal UI

- Built a privacy-first knowledge system as a single Rust binary combining Markdown notes, local AI search, and cited RAG chat with file- and line-level references.
- Implemented hybrid full-text and semantic search with HNSW indexing, backlinks, graph views, and plugin support.

pane — Windows-Native Linux Environment Platform

Tech: Rust, WSL2, XRDP, PowerShell, Windows packaging automation

- Developed a Windows-native control surface to create, launch, repair, and manage Linux desktop environments — shipped an Arch-first MVP packaged as pane.exe with automated release packaging.

SMNTC — Semantic Motion Engine

Tech: TypeScript, Three.js, GLSL, React, Web Components, CLI tooling

- Built a deterministic visual compiler that translates semantic intent into optimized WebGL motion and shader behavior, exposed via React bindings, Web Components, and CLI export targets.

Entangled — Full-Stack Social Web Application with AI Features

Tech: JavaScript, Node.js, Express, MongoDB, REST APIs, Three.js

- Built a full-stack single-page social app (auth, feeds, messaging, notifications, search, uploads) over a REST API backend, with integrated AI chatbot, summarization, and code-explanation features.

EXPERIENCE

Open-Source Contributor, HAGI (Hypercomplex AGI Research)

2026 — Present

github.com/ShmidtS/HAGI | Remote

- Contribute on the experimental branch of an open research architecture exploring Clifford-algebra reasoning in small language models, working across PyTorch, Rust, and Lean4 in a collaborative codebase.

Chairman, MDX Computing Society

Sep 2025 — Present

Middlesex University Dubai | Dubai, UAE

- Lead the university computing society across technical initiatives, events, and community direction.

Committee Chair, BCS Student Chapter

Sep 2025 — Present

BCS, The Chartered Institute for IT | Dubai, UAE

- Chair the BCS student chapter committee, planning and executing student-led technical and professional initiatives.

Student Voice Leader

Oct 2024 — Present

Middlesex University Dubai | Dubai, UAE

- Liaise between students and the programme team on teaching quality, resources, assessments, and student experience, representing consolidated cohort feedback in academic forums.

Customer Service Representative

Oct 2021 — Jan 2022

transcosmos inc. | Debrecen, Hungary

- Resolved customer issues in a structured service environment requiring communication discipline and resilience under pressure.

EDUCATION

BSc (Hons) Information Technology

Sep 2024 — Present

Middlesex University Dubai | Dubai, UAE

A Levels

Completed

Cambridge International School, Dubai | Dubai, UAE

CERTIFICATIONS

- IBM AI Engineering Professional Certificate — Coursera
- Google IT Automation with Python Professional Certificate — Coursera
- Foundations of Project Management — Google / Coursera
- Data Visualization and Dashboards with Excel and Cognos — IBM / Coursera

ADDITIONAL INFORMATION

Languages: English (Native), Urdu (Native), Arabic (Professional Working), Hungarian (Limited Working)

Work Authorization: UAE Golden Visa holder — no sponsorship required

Availability: Internship, junior, contract, freelance, and full-time roles. On-site or hybrid in Dubai / Abu Dhabi, UAE, and remote worldwide. Immediate start.